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Research Paper

The learning expectations of undergraduate nursing students for the flipped *Health Assessment* course: A qualitative study

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ABSTRACT

Objective: This study aimed to investigate the learning expectations of undergraduate nursing students regarding the flipped *Health Assessment* course.
Methods: This descriptive, qualitative study was conducted at a medical university in Fuzhou, Fujian Province, China. An interview outline was designed based on the core dimensions of Expectation Confirmation Theory (expectation sources, expectation content, and expectation importance). Thirty second-year undergraduate nursing students who had completed first-year basic medical courses and were about to take the flipped *Health Assessment* course were interviewed between June and July 2022. Interview data were analyzed using qualitative content analysis.
Results: Five major themes and thirteen subthemes were identified. Theme 1 was expectation sources—perceived learning difficulties from past experiences, which included four subthemes: insufficient autonomous learning ability, confusion regarding learning methods, insufficient engagement in learning, and low professional identity. Theme 2 included knowledge and information expectations, which comprised three subthemes: knowledge to improve professional competence, knowledge to enhance academic competitiveness, and knowledge to boost self-efficacy. Theme 3 comprised logical expectations and included two subthemes: flexible teaching methods and efficient instructional tools. Theme 4, pleasure expectations, included two subthemes: vivid teaching styles and diversified teaching evaluations. Theme 5 comprised professional value expectations and included two subthemes: teachers' responsible professional attitudes and gentle emotional support.
Conclusion: Students' learning expectations originate from perceived learning difficulties, such as insufficient learning autonomy and engagement, confusion about learning methods, and a lack of professional identity. They showed various learning expectations for the flipped *Health Assessment* course, including knowledge and information, logic, pleasure, and professional value expectations.
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What is known?

- The *Health Assessment* course presents learning challenges, yet traditional instructor-centric teaching methods often fail to engage students effectively.
- Flipped courses can address the issue of low engagement among students in traditional teaching. Still, they face significant barriers, such as complex and overwhelming pre-work and uneven in-class participation, leading to student resistance.
- Learning expectations impact students' motivation and engagement and can be used as a target to help fully leverage the advantages and potential of flipped courses.

What is new?

- Students' learning expectations for the flipped *Health Assessment* course originated from perceived learning difficulties from past experiences, such as insufficient learning autonomy and engagement, confusion about learning methods, and a lack of professional identity.
- Students showed various learning expectations for the flipped *Health Assessment* course, including knowledge and information, logic, pleasure, and professional value expectations.

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- Practical suggestions were made based on the results, such as incorporating real-world projects or case studies, personalized learning task sheets, video feedback teaching, interactive digital tools, and multidimensional evaluation.

1. Introduction

Health assessment represents a systematic process designed to identify an individuals' current or potential health issues through a thorough evaluation of their physiological, psychological, and social dimensions. This process is followed by formulating nursing interventions to enhance health outcomes [1]. Health assessment is integral to holistic nursing practices and is recognized as an essential competency required by clinical nurses [1]. Within the framework of nursing education, *Health Assessment* courses facilitate the transition from theoretical knowledge to practical applications for undergraduate students [2]. Nursing students are expected to master health assessment theories and methodologies to cultivate holistic thinking in nursing practice and acquire diagnostic skills that will enable them to develop into proficient professionals with strong ethical standards [3]. *Health Assessment* courses encompass a broad range of content, significant complexity, and increased practical engagement; thus, many students encounter considerable challenges before enrollment [3]. The traditional pedagogical model positions instructors as the primary sources of information, relegating students to passive recipients of knowledge. This dynamic often leads to diminished interest and initiative among students and inadequate retention of knowledge essential for future clinical applications.

The flipped classroom (FC) reverses the traditional sequence of knowledge delivery and assimilation, shifting direct instruction to pre-class, self-paced learning, while reserving in-class time for interactive, applied activities like problem-solving and discussions [4,5]. Nursing courses like *Health Assessment* adopted FC to address gaps in traditional training, such as passive learning and limited clinical-skill practice. A private nursing college in South Korea implemented the FC in one didactic session of a physical assessment course, where students engaged in self-directed pre-class activities via online lecture videos and reading materials, followed by in-class interactive team-based learning and subsequent skills lab practicum. Participants showed significantly increased perceptions of student-centeredness, viewing FC as an acceptable approach to fostering active learning in a supportive environment [6]. The FC integrated with scenario simulation was implemented among second-year undergraduate nursing students, and their knowledge mastery and retention were evaluated in comparison with the traditional FC approach. Regarding knowledge mastery, the FC group achieved higher scores than the FC with scenario simulation (FCSS) group in both total and group learning scores. However, the FCSS group demonstrated better knowledge retention than the FC group [7]. A parallel four-group randomized clinical trial was conducted to compare the effects of FC, gamification, and gamification-integrated flipped learning on performance in client health assessment. Results indicated that FC combined with gamification was associated with greater attractiveness, higher satisfaction, and improved learning outcomes [8].

Despite its advantages, FC faces significant barriers in nursing education. Learners with poor time management or low autonomy often struggle with pre-class preparation, leading to uneven in-class participation. Complex pre-work without adequate support can overwhelm students, hindering readiness for advanced activities. In addition, FC may exacerbate theory-practice disconnects if clinical scenarios are poorly synchronized with pre-class content [9,10]. Therefore, it is necessary to take strategies to fully leverage the advantages and potential of FC to enhance students' motivation and engagement in class.

Expectation Confirmation Theory (ECT) is a well-established theoretical framework that initially sought to elucidate how individuals' post-purchase satisfaction and repurchase intentions are influenced by their pre-purchase expectations and subsequent experiences with products or services. The core premise of ECT is that individuals possess specific expectations regarding a product or service, and their perceptions and attitudes are shaped by the degree to which these expectations are confirmed or disconfirmed [11]. In healthcare education, students invariably hold specific expectations regarding their upcoming courses, referred to as learning expectations, which are formulated based on prior educational experiences and beliefs [12]. The extent to which learning expectations are met directly affects student motivation and satisfaction, making it imperative to address these expectations to enhance learning outcomes [13]. However, research has seldom focused on learning expectations from the perspective of undergraduate nursing students. Further exploration is needed to understand how to enhance students' motivation and satisfaction by managing learning expectations.

Consequently, this study aimed to investigate the learning expectations held by nursing undergraduates regarding the flipped *Health Assessment* course through qualitative research, to bolster students' motivation for learning, optimize educational outcomes, and contribute strategic recommendations for ongoing improvements in teaching practices.

2. Methods

This descriptive, qualitative study was conducted from June to July 2022 at a medical university in Fuzhou, Fujian province, China. Data were obtained from semi-structured individual interviews and analyzed using qualitative content analysis. The reporting of this study adhered to the Consolidated Criteria for Reporting Qualitative Research (COREQ) guidelines [14].

2.1. Participants

Full-time undergraduate nursing students were selected using purposive sampling to ensure diverse representations. Four sampling frames were established based on class (classes 1 and 2) and gender (male and female). Students with different intellectual achievement rankings (upper middle, middle, and lower middle) were evaluated based on their grade point average (GPA) in the first-year medical basic courses. They were then selected from each sampling frame using the random number table method to achieve an equitable distribution. The inclusion criteria for participants were as follows: 1) going to be a sophomore in college; 2) have completed first-year basic courses including *Physiology*, *Pathology*, *Human Morphology*, *Medical Microbiology*, *Biochemistry* and *Molecular Biology*, *Introduction to Nursing*; and 3) were about to take the flipped *Health Assessment* courses. Exclusion criteria included: 1) intentions to withdraw from school or change their major, and 2) experiencing serious illness or significant negative or traumatic events in the past year. The sample size was determined based on the estimated number of participants required to achieve data saturation, defined as the point at which no new issues emerge and the code book begins to stabilize [15].

2.2. Data collection

The ECT provides a robust framework for enhancing understanding of the mechanisms underlying learning expectations among undergraduate nursing students [11]. We designed a

preliminary interview outline that aligns with the ECT's core dimensions as expectation sources (previous learning experience and information), expectation contents (cognitive knowledge and information expectation, psychomotor method and tool expectation, and affective pleasure and professional value expectation), and expectation importance. The preliminary interview outline was refined through pilot interviews with two students. The final interview outline consisted of seven questions, listed as "Can you share with me your experience of studying medical basic courses in the past year, what difficulties or challenges did you encounter ?" "Do you know about *Health Assessment* course, what do you know about it ?" "Do you know about FC, what do you know about it ?" "What are your cognitive (knowledge and information) expectations for theflipped *Health Assessment* course" "What are your psychomotor (method and tool) expectations for the flipped *Health Assessment* course ?" "What are your affective (pleasure and professional value) expectations for the flipped *Health Assessment* course ?" "How critical are these expectations to your learning?"

The researchers made leaflets and distributed them to potential participants in the classroom, fully elucidated the research purpose and procedure, and left contact information. Semi-structured interviews were conducted in a classroom at times preferred by the participants, 30–60 min each time. The interviews included open-ended questions derived from predefined outlines. Interviews involved only the researcher and the participants individually. Each interview was audio-recorded and transcribed verbatim in Mandarin, highlighting key points. The transcripts were subsequently returned to the participants for review and correction. Following the completion of 30 interviews, no new themes emerged. Consequently, the research team conducted two additional interviews and analyzed them to confirm the absence of new data or themes. Thus, data collection was concluded.

2.3. Data analysis

The recordings and interview notes were transcribed within 24 h after each sessions, adhering to thematic analysis methodology [16] using Nvivo 11.0. The interview transcripts were reviewed and coded to develop the initial codebook. Similar codes were subsequently grouped into categories and organized into overarching themes. In addition, the transcripts, codes, categories, and themes were translated into English. A PhD nursing researcher who had studied in an international exchange program for more than one year participated in the back-translation of all study materials and reviewed the transcripts and translations for accuracy. In disputes during the theme extraction process, a consensus was reached through discussion and arbitration by a senior researcher.

2.4. Ethical consideration

Students could sign the informed consent form immediately, or contact the researchers by phone after careful consideration and sign the informed consent form to initiate the enrollment process. All individuals contacted consented to engage in the interviews. This study was exempt from ethical review by Article 3 of the Measures for the Ethical Review of Biomedical Research Involving Humans (National Health and Family Planning Commission, 2016) for the following reasons: 1) the study was conducted via semi-structured interviews that involve no medical interventions or collection of patient medical records; 2) no personally identifiable or medical information was collected; and 3) participants were students rather than patients, and all response were submitted anonymously.

2.5. Qualitative rigor

Several strategies ensured the rigor of the research implementation. First, following systematic training, the principal researcher (first author) mastered semi-structured interview procedures, precautions, and verbal/non-verbal behavior recording methods. Second, all interviews were conducted one-on-one between researchers and interviewees per the interview protocol, with no irrelevant personnel present and no leading questions formulated or posed. Third, interviews and analyses involved systematic transcription and verbatim comprehension, with transcribed data reviewed by multiple researchers at each phase to ensure fidelity. Finally, the supervisor supervised the research process; potential biases were mitigated via reflective practices, while full research team involvement in all analytical stages and critical methodology evaluation enhanced findings' credibility.

3. Results

3.1. General information about the students

The study involved a sample of 30 participants, aged between 17 and 21 years (Appendix A). Through a rigorous and iterative analytical process, the identified codes were categorized into five overarching themes and thirteen subthemes: 1) expectation sources-perceived learning difficulties from past experiences (Insufficient autonomous learning ability, confusion regarding learning methods, insufficient engagement in learning, and low professional identity); 2) knowledge and information expectations (knowledge to improve professional competence, knowledge to enhance academic competitiveness, knowledge to boost self-efficacy); 3) logical Expectations (flexible teaching methods and efficient instructional tools); 4) pleasure expectations (vivid teaching style and diversified teaching evaluations); and 5) professional value expectations (teachers' responsible professional attitudes and gentle emotional support).

3.2. Theme 1: expectation sources-perceived learning difficulties from past experiences

Based on prior study experiences of the basic courses, students reported encountering several learning challenges, including a limited capacity for autonomous learning, confusion regarding effective learning strategies, insufficient engagement in educational activities, and a lack of professional identity.

3.2.1. Low autonomous learning ability

The students interviewed indicated that they primarily engaged with the teacher's lectures during class, and rarely dedicated additional time after class. Some students expressed uncertainty regarding the objectives of these basic medical courses and their relevance to future career opportunities and development. This lack of clarity contributed to diminished motivation to learn, resulting in behaviors such as surfing the Internet or sleeping during class.

> *"The stuff we learnt in these medical basic courses was super boring, and honestly, what was the point? I just showed up for class to get through the exam. As long as I passed the final stage, everything was fine. Sometimes I skipped class and chilled in my dormitory, sleeping or gaming. If the teacher attended, I had to drag myself into class, and it was such a hassle."* (S17)

> *"I do not see the need to spend extra time reviewing what I have learned in the class. I need to pay attention now and then jot down the important things. I was only there to pass the final*

examination. If I flunk it, I will always take it. Additionally, these classes were not as fun as scrolling through my phone. I often found myself browsing the Internet during lectures and could not help it." (S20)

3.2.2. Confusion regarding learning methods

Several respondents indicated that their learning efforts did not yield proportional outcomes. Despite these diligent studies, the results have often been unsatisfactory. Additionally, some respondents stated that basic medical courses were excessively challenging, leading to consistently low learning efficiency. They expected flipped learning to provide effective solutions, especially for professional courses such as *Health Assessment*.

"I felt like some of the stuff we learnt in the medical basic courses was way too confusing. The teacher did not interact with us in the class. I honestly did not know where to begin. Plus, if I got distracted in class, I lost track of what the teacher was saying, and then I felt like I had wasted my time." (S7)

"I have been struggling with my study methods. Even when I put in effort and hours, my grades did not reflect this. I heard that the Health Assessment course was practical. Case studies and simulation scenarios are good ways of learning. Can these be realized in a FC" (S14)

3.2.3. Insufficient engagement in learning

Students struggle with overly challenging courses due to limited opportunities for clarification. Sudden and poorly organized group tasks can lead to disengagement, exacerbating free-riding problems. According to students, inadequate teacher-student communication and poorly designed collaborative activities hinder class engagement. Addressing these issues requires pedagogical flexibility, clear task frameworks, and opportunities for sustained interactions beyond traditional classroom settings.

"These classes were way harder than the others. I often zone out in class because I do not get it. I could not find an appropriate way to ask the teacher questions after class. I learned that Health Assessment courses are complex and difficult to learn. If we have the chance to interact with the teacher outside class, such as learning cases in advance and receiving feedback after class, we may achieve better learning results. I heard that the flipped classroom is adopted in this form; I look forward to it." (S15)

"Sometimes the teacher gave us group assignments during class, suddenly. I felt it was a total waste of time, and we could just let the group leader handle it. As long as I was not a team leader, I did not need much effort; other students might think the same way. If the tasks for each group member were assigned, and everyone was given sufficient time to prepare in advance, this problem could be solved." (S24)

3.2.4. Low professional identity

The student regretted choosing the major without careful consideration, finding last year's basic medical courses full of dull medical knowledge, and doubted if the *Health Assessment* course would be the same. In addition, after one year of basic courses, the students could not distinguish nursing from other medical majors. They expected the upcoming *Health Assessment* course to clarify the connotations and values of nursing partially.

"When I picked this major, I did not think it through. I did not feel the value of this subject from the basic medical courses I learned last year–just a lot of raw and boring medical knowledge. Will the Health Assessment courses be the same? I hope that the content and teaching methods of the Health Assessment course can inspire me to develop a greater sense of professional identity." (S4)

"Although I have studied basic medical courses for a year, I still cannot distinguish nursing from other medical majors. The Health Assessment course will be the first basic nursing professional course I will take, and I hope it can help me understand the connotations and value of nursing." (S30)

3.3. Theme 2: knowledge and information expectations

The students anticipated that acquiring knowledge and information would enable them to navigate the perceived challenges linked to their limited autonomous learning abilities. They expected knowledge to improve their professional competence, enhance their academic competitiveness, and boost their self-efficacy.

3.3.1. Knowledge to improve professional competence

The students recognized a deficit in practical skills, attributing it to insufficient in-class practice and an overemphasis on theory by past teachers. They hoped the flipped *Health Assessment* course would provide more hands-on training opportunities to compensate for this shortcoming. Moreover, they were excited about this course, as it combined theory and practice, which they believed would enhance their professional competence.

"I feel fairly good about my theoretical knowledge, but my practical skills are lacking, probably because I have not practiced enough in class. I wonder if the FC can increase my skill practice opportunities and compensate for this deficiency."(S8)

"I really think having solid practical skills will help me in my future career, but my past teachers focused too much on theory and did not provide enough hands-on training. I hope that the flipped Health Assessment courses will emphasize building those operational skills more." (S16)

"The Health Assessment course is going to be my first professional class. This is different from the basic courses that I took last year. It combines theory with practice and feels more professional. I am super excited to take this course!" (S22)

3.3.2. Knowledge to enhance academic competitiveness

With increasing nursing students and escalating employer expectations, students reported experiencing heightened pressure to secure employment and pursue further education. They expressed concerns that failing to distinguish themselves in academic competition could jeopardize their prospects in the job market.

"Before, all we needed to be a nurse was a college degree, but now many graduates struggle to land good jobs in hospitals. The job market seems to be getting tougher, and I think schools should teach more practical skills to help us stand out academically." (S15)

"I care about academic competitiveness, especially since I plan to continue my studies and get involved in academic work later." (S21)

3.3.3. Knowledge to boost self-efficacy

Students expressed their struggles with basic medical courses. They hoped that the FC would help them overcome in-class distractions. These difficulties imply that the appropriate knowledge and teaching methods in these courses can boost self-efficacy.

> *"These basic medical courses are difficult. I did not seem to get the hang of them, and honestly, I did not think I would ever. I often felt like giving up in class because I could not keep up with the teacher's pace. I felt that there was nothing I could do. I hope the flipped class will be more interesting, and the teacher-student interaction will help me overcome the problem of being distracted in class."* (S17)

> *"Other students seem to remember everything the teacher talks about in class. Am I that slow that I had to study so hard just to get by? Perhaps medicine is not for me. Fingers crossed the Health Assessment course will not be too tough, so that I can learn better."*(S18)

3.4. Theme 3: logical expectations

The students articulated logical expectations to address perceived challenges associated with the uncertainty regarding their learning methodologies. They anticipated that educators would implement adaptable teaching strategies and practical instructional resources in the classroom.

3.4.1. Flexible teaching methods

The teaching style of a storytelling teacher, characterized by clear logic and an emphasis on key points, was recalled, and the students wished that such engaging methods could be applied to a flipped course. Moreover, the students recommended incorporating video feedback into flipped *Health Assessment* course.

> *"You reminded me of a teacher who explained things through stories. His logic was obvious, and he always highlighted key points well, which made me pay close attention during his lectures. If only these fun learning methods could be used in the flipped Health Assessment course"* (S11)

> *"I heard that the Health Assessment course included physical examination procedures. If we watch the teacher demonstrate them, we might not remember them or forget them quickly. I believe we can record our videos of the procedures in advance and upload them to a learning platform. The teacher can comment on and explain them during class, and then practice and consolidate our knowledge based on the teacher's feedback after class. It is a rather interesting process, and it can help us get to grips with all the knowledge and skills we need."* (S14)

3.4.2. Efficient instructional tools

The students suggested that using models was a more effective teaching method than simply showing pictures, as it helped them gain a clearer understanding and reduced their anxiety. They also proposed that creating videos or animations could capture students' attention more effectively than writing on a blackboard and enhance knowledge retention. The personalized learning task list was also deemed an effective instructional tool for the *flipped Health Assessment* course.

> *"If teachers could use models in the Health Assessment course, we would better understand the content, and it would not feel awkward. I think this is a better teaching tool than simply showing pictures."* (S1)

> *"If a teacher can make videos or animations to explain things in the Health Assessment class, it will attract my attention more than just writing on the blackboard, and it will stick with me better."* (S2)

> *"I have heard that the FC requires many pre-class assignments, and just thinking about it makes me feel overwhelmed and stressed. I want a personalized learning task list to help us complete our learning tasks step-by-step. It could also track which areas of knowledge we have acquired and which need improvement."* (S26)

3.5. Theme 4: pleasure expectations

The expectation of pleasure in learning addresses the emotional needs of students, who hope to engage in classes that stimulate their interest, thereby helping them overcome the perceived learning difficulties associated with insufficient engagement.

3.5.1. Vivid teaching style

A vivid teaching style is highlighted by students' emphasis on engagement, energy, and nonverbal communication as critical elements for effective learning. The students collectively underscored that teachers who adopted a friendly, interactive demeanor created a dynamic classroom atmosphere, in contrast to rigid, impersonal instruction. Additionally, energetic delivery was essential for emphasizing key concepts and sustaining focus, whereas passionate lecturing inspired intrinsic motivation and reduced distraction.

> *"If the teacher was friendly and connected with us like a buddy, the class felt way more engaging. Otherwise, it would feel fairly mechanical, and we would not learn as much."* (S4)

> *"I hope that the teacher can make Health Assessment lesson more lively. Discussing key points would be helpful if they raised their voices and used more body language. That way, we would know the important things are being covered and pay more attention."* (S7)

> *"When the teacher was energetic and passionate during lectures, it inspired me. I found it easier to stay focused and not be distracted."*(S19)

3.5.2. Diversified teaching evaluation

Student feedback highlights that integrating regular classroom participation with exam scores can create a more equitable and comprehensive evaluation system. This approach would reduce the pressure on students who struggle with high-stakes testing. By balancing diverse assessment criteria, teachers can foster a supportive learning environment that values academic rigor and practical skill development, ultimately enhancing student engagement and the long-term retention of course material.

> *"I have heard that the Health Assessment course has a lot of professional knowledge. Involved, but also focused on practical skills. It does not seem fair to judge someone's knowledge and abilities based solely on the final exam."* (S1)

> *"If we could emphasize regular classroom participation and the final exam scores in the Health Assessment class, those who did not test well would feel much less pressure, which would benefit us."* (S21)

3.6. Theme 5: professional value expectations

Professional value expectations pertain to students' anticipations regarding personal ethics and instructors' values. They seek to acquire knowledge from their teachers and observe their professional demeanors, ethical standards, and overall character through their actions and words. Meeting these expectations was instrumental in strengthening students' professional identities.

3.6.1. Teachers' responsible professional attitude

Interviewees conveyed their belief that respected educators must possess high professional standards and a responsible attitude, characterizing teachers as highly charismatic and engaging, and motivating students to learn in class.

> *"I had this one teacher who highlighted the importance of humanistic care in nursing. She seemed passionate and responsible for the subject, and I wanted her to present the Health Assessment course."* (S8)

> "A teacher in my operating class was pretty rough and did not care about patient privacy. I thought that was kind of off, but she was a teacher, so I did not feel like I could say anything."(S23)

3.6.2. Teachers' gentle emotional support

The students mentioned that the teacher was knowledgeable and treated them as equals when answering questions, showing respect and friendliness, which provided them with a positive emotional experience. Students also pointed out that some great teachers seemed cold and distant, and hoped that the flipped course would bring teachers and students closer together. This implies that gentle emotional support from teachers, such as approachability, is crucial to students' learning motivation.

> *"I think our teacher knew his stuff well. When asked questions, he answered us nicely and treated us as equals while delivering the correct answer."* (S12)

> *"Even when some teachers were great at teaching, they came off as fairly cold and distant. I hope the FC will bring teachers and students closer together, and I will be more motivated to take this course."* (S21)

4. Discussion

In this study, we interviewed 30 undergraduate nursing students to investigate their learning expectations regarding flipped *Health Assessment* course, which originated from their learning experiences in basic medical courses they previously undertook. Perceived learning difficulties, including insufficient autonomous learning ability, confusion regarding learning methods, insufficient engagement in learning activities, and a lack of professional identity, significantly impacted study satisfaction and shaped students' learning expectations for the forthcoming flipped *Health Assessment* course, such as knowledge and information expectations, logical expectations, pleasure expectations, and professional value expectations.

4.1. Learning expectations originating from perceived learning difficulties for medical basic courses

As an essential approach for lifelong learning among nursing professionals, autonomous learning empowers students to enhance their academic performance, clinical skills, and adaptability in clinical settings [17]. However, undergraduate nursing students often experience insufficient autonomy in their learning. Additionally, many students reported struggles related to confusion over effective learning methods and expressed uncertainty about how to apply theoretical knowledge in practical care scenarios. Learning engagement refers to students' levels of interest, involvement, and investment in their learning activities [18]. This significantly influences learning outcomes and academic achievement [19]. Understanding the factors that affect learning engagement is essential, particularly from the learner's perspective. However, research in this area remains scarce [20]. Professional identity refers to how individuals perceive themselves within a specific profession or occupational role. A strong professional identity is correlated with higher academic achievement among college students [21]. Therefore, nurturing professional identity can enhance academic success in nursing education by providing a foundation for targeted interventions to foster personal and educational growth [18].

4.2. Learning expectations for the Health Assessment flipped course

The *Health Assessment* course constitutes the inaugural nursing course students undertake, which is paramount in shaping their nursing careers. The findings highlighted various dimensions of expectations among students, such as knowledge and information expectations, logical expectations, pleasure expectations, and professional value expectations.

Knowledge and information expectations reflect students' desires regarding learning content. Meeting these knowledge-related needs, such as acquiring knowledge to improve professional competence, enhance academic competitiveness, and boost self-efficacy, can help students overcome perceived challenges related to insufficient autonomous learning abilities. Insufficient autonomous learning often stems from students' perceptions of irrelevance, as they may fail to see how classroom instruction connects to their future careers. When knowledge acquisition is directly linked to professional development, students develop clear and tangible learning purposes. This sense of purpose motivates them to take the initiative, such as seeking additional resources, practicing skills independently, or exploring advanced topics, because they recognize the value of their efforts [22]. Incorporating real-world projects or case studies into flipped lessons can serve as an effective strategy for bridging theoretical knowledge with practical applications. Low levels of autonomous learning can result from unclear goals or inadequate feedback. When motivated to improve academically, students are more likely to seek resources, practice self-regulation, and persist despite challenges. Academic competitiveness fosters a goal-oriented mindset that supports autonomous learning [23]. Providing timely and specific feedback on assignments via FC online learning platforms encourages students to reflect on their progress, identify their strengths and weaknesses, and become more motivated to learn independently. Self-efficacy—the belief in one's ability to succeed—is essential to autonomous learning. Students with low self-efficacy often avoid challenging tasks or give up easily because of doubts about their capabilities [22]. As knowledge acquisition leads to mastery experiences, student self-efficacy improves. This growing confidence makes them more willing to tackle complex tasks, persist despite setbacks, and engage in independent learning.

Logical expectations refer to students' hopes that the teacher's pedagogical methods, ideas, and pacing are aligned with cognitive and psychological habits. Meeting logical expectations, such as employing flexible teaching methods and efficient instructional

tools, can help students overcome perceived difficulties related to confusion about learning approaches. Students are more likely to retain information when presented in a manner that aligns with their learning styles [24,25]. Combining visual, auditory, and kinesthetic activities can effectively cater to the diverse ways students process and internalize knowledge. Instructional tools such as diagrams, infographics, and video tutorials are particularly effective for visual learners. Auditory learners benefited the most from group discussions, podcasts, or teacher-led verbal explanations. Kinesthetic learners gain greater understanding through hands-on experiments or role-playing exercises. Additionally, integrating interactive digital tools into instruction enhances student participation, makes learning more engaging, and reduces feelings of boredom or confusion [26,27]. In addition, the interviewed students suggested personalized learning task sheets to help them organize pre-class tasks, master key knowledge during class, and obtain feedback on their learning outcomes. A video feedback teaching method was also recommended, in which students recorded videos before class, teachers provided feedback, students evaluated each other in groups during class, and students summarized and shared their experiences after class. This will help students master the basic operations of health assessment throughout the process and cultivate their independent learning abilities.

Pleasure expectation represents an emotional need among students, facilitating a relaxed and positive emotional responses. The fulfillment of pleasure expectations can help students overcome the perceived learning difficulties associated with insufficient engagement. Learning engagement is a critical mediating variable that links the educational atmosphere to professional commitment. A positive educational environment correlated with elevated levels of learning engagement and professional commitment [28]. In order to address concerns about students' perceived usability, immersive virtual reality simulations may be a valuable complementary modality for acquiring fundamental nursing skills [29]. Integrating game-based learning with self-regulated learning strategies can enhance student learning performance, self-efficacy, and motivation [30]. In addition, the students anticipated diverse teaching evaluations to meet their expectations of pleasure. A mobile-based multimedia nursing competency evaluation system was developed to assess student performance during nursing practice. This system was more realistic, interactive, and satisfactory than the traditional text-based mobile assessments [31]. The characteristics of FC, such as high interactivity and interest, as well as multidimensional evaluation, can provide students with rich learning experiences to compensate for and overcome their deficiency in insufficient learning engagement.

Professional value expectation refers to students' anticipation of aligning teachers' statements and actions with socially recognized moral norms and values. This expectation represents a demand for ethical conduct. Students communicated their professional expectations regarding the *Health Assessment* course. They viewed instructors as role models and anticipated that teachers would prioritize humanistic care and patient privacy. Moreover, they expected deeper interactions with teachers to foster increased motivation for learning and engagement. FC helps achieve this goal by growing teacher-student interactions, such as pre-class task assignments and timely feedback after class. Meeting professional value expectations also fosters positive classroom engagement and learning dedication, and improves study performance to effectively prepare students for sustained engagement in the workforce [32–34].

4.3. Limitations and suggestions for future research

Certain limitations must be acknowledged. We only did interviews at one university, limiting the generalizability of the results. Future research could benefit from incorporating students from multiple institutions or varied contexts. Additionally, our study exclusively gathered the perspectives of undergraduate students; incorporating the views of master's degree students in future investigations could enhance the findings. Furthermore, we did not employ every aspect of ECT to elucidate the mechanisms by which students developed their learning expectations; further exploration of the contributions of ECT to study design and result interpretation is warranted. The extent of student satisfaction and expectation confirmation regarding the *Health Assessment* course can be further elucidated upon course completion in the future.

5. Conclusion

Students' learning expectations originate from perceived learning difficulties from past experiences, such as insufficient learning autonomy and engagement, confusion about learning methods, and a lack of professional identity. They showed various learning expectations for the flipped *Health Assessment* course, including knowledge and information, logic, pleasure, and professional value expectations.

CRediT authorship contribution statement

Yixin Luo: Data Curation, Methodology, Formal analysis, Writing–original draft. **Jingjing You:** Investigation, Data Curation, Methodology. **Yimin Chen:** Methodology, Investigation. **Meijing Chen:** Methodology, Investigation. **Jiayuan Zhuang:** Methodology, Investigation. **Fangli Xu:** Data Curation. **Rong Hu:** Project administration.

Data availability statement

The datasets generated during and/or analyzed during the current study are available from the corresponding author upon reasonable request.

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Declaration of competing interest

The authors declare there is no conflict of interest.

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Appendices. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ijnss.2025.08.001.

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